

```
labuser@lab-ubuntu-01:~$ sudo apt install ros-noetic-slam-gmapping
Reading package lists... Done
Building dependency tree
Reading state information... Done
The following additional packages will be installed:
  ros-noetic-gmapping ros-noetic-openslam-gmapping
The following NEW packages will be installed:
  ros-noetic-gmapping ros-noetic-openslam-gmapping ros-noetic-slam-gmapping
0 upgraded, 3 newly installed, 0 to remove and 10 not upgraded.
Need to get 427 kB of archives.
After this operation, 2.532 kB of additional disk space will be used.
Do you want to continue? [Y/n] y
Get:1 http://packages.ros.org/ros/ubuntu focal/main amd64 ros-noetic-openslam-gmapping
amd64 0.2.1-1focal.20250426.005832 [154 kB]
Get:2 http://packages.ros.org/ros/ubuntu focal/main amd64 ros-noetic-gmapping amd64
1.4.2-1focal.20250520.001832 [272 kB]
Get:3 http://packages.ros.org/ros/ubuntu focal/main amd64 ros-noetic-slam-gmapping amd64
1.4.2-1focal.20250520.022155 [2.060 B]
Fetched 427 kB in 2s (249 kB/s)
Selecting previously unselected package ros-noetic-openslam-gmapping.
(Reading database ... 323055 files and directories currently installed.)
Preparing to unpack
.../ros-noetic-openslam-gmapping_0.2.1-1focal.20250426.005832_amd64.deb ...
Unpacking ros-noetic-openslam-gmapping (0.2.1-1focal.20250426.005832) ...
Selecting previously unselected package ros-noetic-gmapping.
Preparing to unpack .../ros-noetic-gmapping_1.4.2-1focal.20250520.001832_amd64.deb ...
Unpacking ros-noetic-gmapping (1.4.2-1focal.20250520.001832) ...
Selecting previously unselected package ros-noetic-slam-gmapping.
Preparing to unpack .../ros-noetic-slam-gmapping_1.4.2-1focal.20250520.022155_amd64.deb
...
Unpacking ros-noetic-slam-gmapping (1.4.2-1focal.20250520.022155) ...
Setting up ros-noetic-openslam-gmapping (0.2.1-1focal.20250426.005832) ...
Setting up ros-noetic-gmapping (1.4.2-1focal.20250520.001832) ...
Setting up ros-noetic-slam-gmapping (1.4.2-1focal.20250520.022155) ...
labuser@lab-ubuntu-01:~$
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...
Unpacking ros-noetic-slam-gmapping (1.4.2-1focal.20250520.022155) ...
Setting up ros-noetic-openslam-gmapping (0.2.1-1focal.20250426.005832) ...
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Setting up ros-noetic-slam-gmapping (1.4.2-1focal.20250520.022155) ...
labuser@lab-ubuntu-01:~$ rostopic list | grep odom
ERROR: Unable to communicate with master!
labuser@lab-ubuntu-01:~$ rostopic list | grep odom
/base/odometry_controller/odometry
labuser@lab-ubuntu-01:~$ rostopic echo -n 1 /odom
WARNING: topic [/odom] does not appear to be published yet
^Clabuser@lab-ubuntu-01:~$ roslaunch gmapping slam_gmapping scan:=/scan_unified
Warning: TF_REPEATED_DATA ignoring data with redundant timestamp for frame odom
(parent map) at time 41.535000 according to authority unknown_publisher
        at line 277 in /tmp/binarydeb/ros-noetic-tf2-0.7.10/src/buffer_core.cpp
```

[WARN] [1783022342.496797939, 56.435000000]: MessageFilter [target=odom]: Dropped 100.00% of messages so far. Please turn the [ros.gmapping.message_filter] rosconsole logger to DEBUG for more information.

RESULTS: for all Frames

Frames:

Frame: /sensorring_cam3d_back_depth_frame published by unknown_publisher(static) Average Delay: 0 Max Delay: 0

Frame: /sensorring_cam3d_back_depth_optical_frame published by unknown_publisher(static) Average Delay: 0 Max Delay: 0

Frame: /sensorring_cam3d_back_frame published by unknown_publisher(static) Average Delay: 0 Max Delay: 0

Frame: /sensorring_cam3d_back_optical_frame published by unknown_publisher(static) Average Delay: 0 Max Delay: 0

Frame: /sensorring_cam3d_back_rgb_frame published by unknown_publisher(static) Average Delay: 0 Max Delay: 0

Frame: /sensorring_cam3d_back_rgb_optical_frame published by unknown_publisher(static) Average Delay: 0 Max Delay: 0

Frame: /torso_cam3d_left_depth_frame published by unknown_publisher(static) Average Delay: 0 Max Delay: 0

Frame: /torso_cam3d_left_depth_optical_frame published by unknown_publisher(static) Average Delay: 0 Max Delay: 0

Frame: /torso_cam3d_left_frame published by unknown_publisher(static) Average Delay: 0 Max Delay: 0

Frame: /torso_cam3d_left_optical_frame published by unknown_publisher(static) Average Delay: 0 Max Delay: 0

Frame: /torso_cam3d_left_rgb_frame published by unknown_publisher(static) Average Delay: 0 Max Delay: 0

Frame: /torso_cam3d_left_rgb_optical_frame published by unknown_publisher(static) Average Delay: 0 Max Delay: 0

Frame: /torso_cam3d_right_depth_frame published by unknown_publisher(static) Average Delay: 0 Max Delay: 0

Frame: /torso_cam3d_right_depth_optical_frame published by unknown_publisher(static) Average Delay: 0 Max Delay: 0

Frame: /torso_cam3d_right_frame published by unknown_publisher(static) Average Delay: 0 Max Delay: 0

Frame: /torso_cam3d_right_optical_frame published by unknown_publisher(static) Average Delay: 0 Max Delay: 0

Frame: /torso_cam3d_right_rgb_frame published by unknown_publisher(static) Average Delay: 0 Max Delay: 0

Frame: /torso_cam3d_right_rgb_optical_frame published by unknown_publisher(static) Average Delay: 0 Max Delay: 0

Frame: arm_left_1_link published by unknown_publisher Average Delay: 0.000298246 Max Delay: 0.001

Frame: arm_left_2_link published by unknown_publisher Average Delay: 0.000315789 Max Delay: 0.001

Frame: arm_left_3_link published by unknown_publisher Average Delay: 0.000333333 Max Delay: 0.001

Frame: arm_left_4_link published by unknown_publisher Average Delay: 0.000350877 Max Delay: 0.001

Frame: arm_left_5_link published by unknown_publisher Average Delay: 0.000350877 Max Delay: 0.001

Frame: arm_left_6_link published by unknown_publisher Average Delay: 0.000350877 Max Delay: 0.001

Frame: arm_left_7_link published by unknown_publisher Average Delay: 0.000350877 Max Delay: 0.001

Frame: arm_left_7_target published by unknown_publisher Average Delay: 2.6087e-05 Max Delay: 0.001

Frame: arm_left_base_link published by unknown_publisher(static) Average Delay: 0 Max Delay: 0

Frame: arm_right_1_link published by unknown_publisher Average Delay: 0.000350877 Max Delay: 0.001

Frame: arm_right_2_link published by unknown_publisher Average Delay: 0.000350877 Max Delay: 0.001

Frame: arm_right_3_link published by unknown_publisher Average Delay: 0.000368421 Max Delay: 0.001

Frame: arm_right_4_link published by unknown_publisher Average Delay: 0.000368421 Max Delay: 0.001

Frame: arm_right_5_link published by unknown_publisher Average Delay: 0.000368421 Max Delay: 0.001

Frame: arm_right_6_link published by unknown_publisher Average Delay: 0.000368421 Max Delay: 0.001

Frame: arm_right_7_link published by unknown_publisher Average Delay: 0.000368421 Max Delay: 0.001

Frame: arm_right_7_target published by unknown_publisher Average Delay: 3.47826e-05 Max Delay: 0.001

Frame: arm_right_base_link published by unknown_publisher(static) Average Delay: 0 Max Delay: 0

Frame: b_caster_r_wheel_link published by unknown_publisher Average Delay: 0.000368421 Max Delay: 0.001

Frame: b_caster_rotation_link published by unknown_publisher Average Delay: 0.000368421 Max Delay: 0.001

Frame: base_charger_link published by unknown_publisher(static) Average Delay: 0 Max Delay: 0

Frame: base_chassis_link published by unknown_publisher(static) Average Delay: 0 Max Delay: 0

Frame: base_footprint published by unknown_publisher Average Delay: 5.17241e-05 Max Delay: 0.001

Frame: base_laser_front_link published by unknown_publisher(static) Average Delay: 0 Max Delay: 0

Frame: base_laser_left_link published by unknown_publisher(static) Average Delay: 0 Max Delay: 0

Frame: base_laser_right_link published by unknown_publisher(static) Average Delay: 0 Max Delay: 0

Frame: base_link published by unknown_publisher(static) Average Delay: 0 Max Delay: 0

Frame: fl_caster_r_wheel_link published by unknown_publisher Average Delay: 0.000368421 Max Delay: 0.001

Frame: fl_caster_rotation_link published by unknown_publisher Average Delay: 0.000368421 Max Delay: 0.001

Frame: fr_caster_r_wheel_link published by unknown_publisher Average Delay: 0.000368421 Max Delay: 0.001

Frame: fr_caster_rotation_link published by unknown_publisher Average Delay: 0.000368421 Max Delay: 0.001

Frame: gripper_left_base_link published by unknown_publisher(static) Average Delay: 0 Max Delay: 0

Frame: gripper_left_camera_link published by unknown_publisher(static) Average Delay: 0 Max Delay: 0

Frame: gripper_left_finger_1_link published by unknown_publisher Average Delay: 0.000368421 Max Delay: 0.001

Frame: gripper_left_finger_2_link published by unknown_publisher Average Delay: 0.000368421 Max Delay: 0.001

Frame: gripper_left_grasp_link published by unknown_publisher(static) Average Delay: 0 Max Delay: 0

Frame: gripper_left_palm_link published by unknown_publisher(static) Average Delay: 0 Max Delay: 0

Frame: gripper_right_base_link published by unknown_publisher(static) Average Delay: 0 Max Delay: 0

Frame: gripper_right_camera_link published by unknown_publisher(static) Average Delay: 0 Max Delay: 0

Frame: gripper_right_finger_1_link published by unknown_publisher Average Delay: 0.000368421 Max Delay: 0.001

Frame: gripper_right_finger_2_link published by unknown_publisher Average Delay: 0.000368421 Max Delay: 0.001

Frame: gripper_right_grasp_link published by unknown_publisher(static) Average Delay: 0 Max Delay: 0

Frame: gripper_right_palm_link published by unknown_publisher(static) Average Delay: 0 Max Delay: 0

Frame: head_1_link published by unknown_publisher Average Delay: 0.000385965 Max Delay: 0.001
Frame: head_2_link published by unknown_publisher Average Delay: 0.000385965 Max Delay: 0.001
Frame: head_3_link published by unknown_publisher Average Delay: 0.000403509 Max Delay: 0.001
Frame: head_base_link published by unknown_publisher(static) Average Delay: 0 Max Delay: 0
Frame: head_cam_frame published by unknown_publisher(static) Average Delay: 0 Max Delay: 0
Frame: head_cam_link published by unknown_publisher(static) Average Delay: 0 Max Delay: 0
Frame: head_cam_optical_frame published by unknown_publisher(static) Average Delay: 0 Max Delay: 0
Frame: head_center_link published by unknown_publisher(static) Average Delay: 0 Max Delay: 0
Frame: head_center_target published by unknown_publisher Average Delay: 3.44828e-05 Max Delay: 0.001
Frame: sensorring_base_link published by unknown_publisher(static) Average Delay: 0 Max Delay: 0
Frame: sensorring_cam3d_back_link published by unknown_publisher(static) Average Delay: 0 Max Delay: 0
Frame: sensorring_cam3d_front_frame published by unknown_publisher(static) Average Delay: 0 Max Delay: 0
Frame: sensorring_cam3d_front_link published by unknown_publisher(static) Average Delay: 0 Max Delay: 0
Frame: sensorring_cam3d_front_optical_frame published by unknown_publisher(static) Average Delay: 0 Max Delay: 0
Frame: sensorring_link published by unknown_publisher Average Delay: 0.000403509 Max Delay: 0.001
Frame: torso_1_link published by unknown_publisher(static) Average Delay: 0 Max Delay: 0
Frame: torso_2_link published by unknown_publisher(static) Average Delay: 0 Max Delay: 0
Frame: torso_3_link published by unknown_publisher(static) Average Delay: 0 Max Delay: 0
Frame: torso_base_link published by unknown_publisher(static) Average Delay: 0 Max Delay: 0
Frame: torso_cam3d_down_link published by unknown_publisher(static) Average Delay: 0 Max Delay: 0
Frame: torso_cam3d_left_link published by unknown_publisher(static) Average Delay: 0 Max Delay: 0
Frame: torso_cam3d_right_link published by unknown_publisher(static) Average Delay: 0 Max Delay: 0
Frame: torso_center_link published by unknown_publisher(static) Average Delay: 0 Max Delay: 0

All Broadcasters:

Node: unknown_publisher 352.89 Hz, Average Delay: 8.10883e-05 Max Delay: 0.001
Node: unknown_publisher(static) 3800 Hz, Average Delay: 0 Max Delay: 0

```
^Clabuser@lab-ubuntu-01:~$ rosrun tf view_frames
WARNING: tf view_frames is deprecated. Use tf2_tools view_frames.py instead.
Listening to /tf for 5.0 seconds
Done Listening
frames.pdf generated
labuser@lab-ubuntu-01:~$ rostopic echo -n 1 /base/odometry_controller/odometry
header:
  seq: 12066
  stamp:
    secs: 241
    nsecs: 732000000
  frame_id: "odom_combined"
child_frame_id: "base_footprint"
pose:
  pose:
    position:
      x: 0.0041750250302242005
      y: 0.00013914067630293122
      z: 0.0
    orientation:
      x: 0.0
      y: 0.0
      z: -0.00026473950870399776
      w: 0.99999999649564957
  covariance: [0.1, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.1, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.1, 0.0, 0.0, 0.0,
0.0, 0.0, 0.0, 0.1, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.1, 0.0, 0.0, 0.0, 0.0, 0.0, 0.1]
twist:
  twist:
    linear:
      x: 2.101187703898925e-09
      y: 9.192895189231257e-15
      z: 0.0
    angular:
      x: 0.0
      y: 0.0
      z: 6.40755778418553e-12
  covariance: [0.1, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.1, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.1, 0.0, 0.0, 0.0,
0.0, 0.0, 0.0, 0.1, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.1, 0.0, 0.0, 0.0, 0.0, 0.0, 0.1]
---
labuser@lab-ubuntu-01:~$
```

```
labuser@lab-ubuntu-01:~$ rostopic echo -n 1 /scan_unified/header
```

```
seq: 3359
```

```
stamp:
```

```
  secs: 284
```

```
  nsecs: 629000000
```

```
frame_id: "base_link"
```

```
---
```

```
labuser@lab-ubuntu-01:~$ rosrn tf tf_echo odom_combined base_footprint
```

```
At time 293.572
```

```
- Translation: [0.005, 0.000, 0.000]
```

```
- Rotation: in Quaternion [0.000, 0.000, -0.000, 1.000]
```

```
          in RPY (radian) [0.000, 0.000, -0.001]
```

```
          in RPY (degree) [0.000, 0.000, -0.029]
```

```
At time 293.572
```

```
- Translation: [0.005, 0.000, 0.000]
```

```
- Rotation: in Quaternion [0.000, 0.000, -0.000, 1.000]
```

```
          in RPY (radian) [0.000, 0.000, -0.001]
```

```
          in RPY (degree) [0.000, 0.000, -0.029]
```

```
At time 294.572
```

```
- Translation: [0.005, 0.000, 0.000]
```

```
- Rotation: in Quaternion [0.000, 0.000, -0.000, 1.000]
```

```
          in RPY (radian) [0.000, 0.000, -0.001]
```

```
          in RPY (degree) [0.000, 0.000, -0.029]
```

```
At time 295.572
```

```
- Translation: [0.005, 0.000, 0.000]
```

```
- Rotation: in Quaternion [0.000, 0.000, -0.000, 1.000]
```

```
          in RPY (radian) [0.000, 0.000, -0.001]
```

```
          in RPY (degree) [0.000, 0.000, -0.030]
```

```
^Clabuser@lab-ubuntu-01:~$
```

```
labuser@lab-ubuntu-01:~$ rostopic list | grep map
```

```
/collision_velocity_filter/anti_collision_costmap/costmap
```

```
/collision_velocity_filter/anti_collision_costmap/costmap_updates
```

```
/collision_velocity_filter/anti_collision_costmap/footprint
```

```
/collision_velocity_filter/anti_collision_costmap/obstacle_layer_0/parameter_descriptions
```

```
/collision_velocity_filter/anti_collision_costmap/obstacle_layer_0/parameter_updates
```

```
/collision_velocity_filter/anti_collision_costmap/parameter_descriptions
```



```
/collision_velocity_filter/anti_collision_costmap/parameter_updates
```

```
/map
```

```
/map_metadata
```

```
/slam_gmapping/entropy
```

```
labuser@lab-ubuntu-01:~$ rostopic hz /map
```

```
subscribed to [/map]
```

```
WARNING: may be using simulated time
```

```
no new messages
```

```
no new messages
```

```
no new messages
```

```
no new messages
```

```
no new messages
```

```
no new messages
```

```
no new messages
```

```
no new messages
```

```
average rate: 0.184
```

```
min: 5.423s max: 5.423s std dev: 0.00000s window: 2
```

```
no new messages
```

```
no new messages
```

```
no new messages
```

```
no new messages
```

```
no new messages
```

```
no new messages
```

```
^Cno new messages
```

```
labuser@lab-ubuntu-01:~$
```

```
labuser@lab-ubuntu-01:~$ rostopic hz /scan_unified
```

```
subscribed to [/scan_unified]
```

```
WARNING: may be using simulated time
```

```
average rate: 11.925
```

```
min: 0.082s max: 0.085s std dev: 0.00099s window: 8
```

```
average rate: 11.905
```

```
min: 0.082s max: 0.086s std dev: 0.00100s window: 15
```

```
average rate: 11.898
```

```
min: 0.082s max: 0.086s std dev: 0.00090s window: 22
```

```
^Caverage rate: 11.905
```

```
min: 0.082s max: 0.086s std dev: 0.00090s window: 23
```

```
labuser@lab-ubuntu-01:~$
```

```
labuser@lab-ubuntu-01:~$ rosrn tf tf_echo odom_combined base_link
```

```
Warning: TF_REPEATED_DATA ignoring data with redundant timestamp for frame
```

```
arm_left_7_target (parent odom_combined) at time 647.882000 according to authority
```

```
unknown_publisher
```

```
at line 277 in /tmp/binarydeb/ros-noetic-tf2-0.7.10/src/buffer_core.cpp
```

Warning: TF_REPEATED_DATA ignoring data with redundant timestamp for frame arm_right_7_target (parent odom_combined) at time 647.882000 according to authority unknown_publisher

at line 277 in /tmp/binarydeb/ros-noetic-tf2-0.7.10/src/buffer_core.cpp

At time 647.883

- Translation: [2.309, -0.373, 0.000]
- Rotation: in Quaternion [0.000, 0.000, 0.924, 0.381]
in RPY (radian) [0.000, -0.000, 2.359]
in RPY (degree) [0.000, -0.000, 135.179]

At time 647.883

- Translation: [2.309, -0.373, 0.000]
- Rotation: in Quaternion [0.000, 0.000, 0.924, 0.381]
in RPY (radian) [0.000, -0.000, 2.359]
in RPY (degree) [0.000, -0.000, 135.179]

At time 648.872

- Translation: [2.309, -0.373, 0.000]
- Rotation: in Quaternion [0.000, 0.000, 0.924, 0.381]
in RPY (radian) [0.000, -0.000, 2.359]
in RPY (degree) [0.000, -0.000, 135.177]

^Clabuser@lab-ubuntu-01:~\$

labuser@lab-ubuntu-01:~\$ rosrn gmapping slam_gmapping \

> scan:=/scan_unified \

> _base_frame:=base_link \

> _odom_frame:=odom_combined

Warning: TF_REPEATED_DATA ignoring data with redundant timestamp for frame odom_combined (parent map) at time 321.213000 according to authority unknown_publisher
at line 277 in /tmp/binarydeb/ros-noetic-tf2-0.7.10/src/buffer_core.cpp

[INFO] [1783022763.415306883, 321.173000000]: Laser is mounted upwards.

-maxUrange 29.49 -maxUrange 29.49 -sigma 0.05 -kernelSize 1 -lstep 0.05 -lobGain 3

-astep 0.05

-srr 0.1 -srt 0.2 -str 0.1 -stt 0.2

-linearUpdate 1 -angularUpdate 0.5 -resampleThreshold 0.5

-xmin -100 -xmax 100 -ymin -100 -ymax 100 -delta 0.05 -particles 30

[INFO] [1783022763.425432482, 321.179000000]: Initialization complete

update frame 0

update ld=0 ad=0

Laser Pose= 0.00514222 0.000138636 -0.000486389

m_count 0

Registering First Scan

update frame 1756
update Id=0.466146 ad=0.503847
Laser Pose= 0.453991 0.10328 0.498417
m_count 1
Average Scan Matching Score=665.919
neff= 29.958
Registering Scans:Done
update frame 1771
update Id=0.476514 ad=0.541278
Laser Pose= 0.795488 0.42786 1.0397
m_count 2
Average Scan Matching Score=673.488
neff= 29.8908
Registering Scans:Done
update frame 1779
update Id=0.338399 ad=0.578973
Laser Pose= 0.881561 0.75029 1.61867
m_count 3
Average Scan Matching Score=646.528
neff= 29.7127
Registering Scans:Done
update frame 1785
update Id=0.252013 ad=0.501232
Laser Pose= 0.808591 0.988842 2.1199
m_count 4
Average Scan Matching Score=674.948
neff= 29.715
Registering Scans:Done
update frame 1791
update Id=0.252416 ad=0.504507
Laser Pose= 0.629369 1.16286 2.62441
m_count 5
Average Scan Matching Score=659.468
neff= 29.8793
Registering Scans:Done
update frame 1798
update Id=0.292878 ad=0.583934
Laser Pose= 0.347947 1.22764 -3.07484
m_count 6
Average Scan Matching Score=650.686
neff= 29.8801
Registering Scans:Done
update frame 1804
update Id=0.251581 ad=0.503517

Laser Pose= 0.111442 1.14968 -2.57133
m_count 7
Average Scan Matching Score=661.146
neff= 29.8093
Registering Scans:Done
update frame 1810
update Id=0.251879 ad=0.504113
Laser Pose= -0.0582887 0.967087 -2.06721
m_count 8
Average Scan Matching Score=663.02
neff= 29.5861
Registering Scans:Done
update frame 1816
update Id=0.252387 ad=0.505146
Laser Pose= -0.118727 0.724725 -1.56207
m_count 9
Average Scan Matching Score=674.035
neff= 29.7189
Registering Scans:Done
update frame 1823
update Id=0.240129 ad=0.503815
Laser Pose= -0.0604071 0.494083 -1.05825
m_count 10
Average Scan Matching Score=677.295
neff= 29.6607
Registering Scans:Done
update frame 1877
update Id=1.03869 ad=0.161129
Laser Pose= 0.579052 -0.322488 -0.897519
m_count 11
Average Scan Matching Score=688.331
neff= 29.6613
Registering Scans:Done
update frame 1901
update Id=1.00756 ad=0.000923314
Laser Pose= 1.20701 -1.11042 -0.897953
m_count 12
Average Scan Matching Score=303.72
neff= 15.2112
Registering Scans:Done
update frame 1949
update Id=0.899947 ad=0.535063
Laser Pose= 1.84941 -1.72255 -0.362891
m_count 13

Average Scan Matching Score=139.661

neff= 1.99993

*****RESAMPLE*****

Deleting Nodes: 0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 19 20 21 22 23 24 25 26 27 29

Done

Deleting old particles...Done

Copying Particles and Registering scans... Done

update frame 2030

update Id=0.411706 ad=0.50245

Laser Pose= 2.22006 -1.89021 -0.570219

m_count 14

Average Scan Matching Score=418.485

neff= 23.7786

Registering Scans:Done

update frame 2044

update Id=0.00110368 ad=0.511935

Laser Pose= 2.22047 -1.88968 -0.0582844

m_count 15

Average Scan Matching Score=648.117

neff= 23.8902

Registering Scans:Done

update frame 2068

update Id=0.000672733 ad=0.512396

Laser Pose= 2.22022 -1.88918 -0.534185

m_count 16

Average Scan Matching Score=666.141

neff= 24.0786

Registering Scans:Done

update frame 2097

update Id=0.141173 ad=0.503309

Laser Pose= 2.14661 -1.76902 -1.03746

m_count 17

Average Scan Matching Score=669.765

neff= 23.9622

Registering Scans:Done

update frame 2108

update Id=0.44948 ad=0.530207

Laser Pose= 2.01927 -1.34324 -1.56767

m_count 18

Average Scan Matching Score=669.402

neff= 24.0693

Registering Scans:Done

update frame 2115

update Id=0.296335 ad=0.537573

Laser Pose= 2.09344 -1.05996 -2.10524

m_count 19

Average Scan Matching Score=671.113

neff= 23.7535

Registering Scans:Done

update frame 2121

update Id=0.252544 ad=0.503729

Laser Pose= 2.27021 -0.883232 -2.60897

m_count 20

Average Scan Matching Score=673.51

neff= 23.6969

Registering Scans:Done

update frame 2127

update Id=0.251897 ad=0.503851

Laser Pose= 2.50973 -0.814038 -3.11282

m_count 21

Average Scan Matching Score=674.137

neff= 23.4267

Registering Scans:Done

update frame 2134

update Id=0.280673 ad=0.566595

Laser Pose= 2.77795 -0.88323 2.60377

m_count 22

Average Scan Matching Score=653.262

neff= 23.1571

Registering Scans:Done

update frame 3123

update Id=0.313886 ad=0.511155

Laser Pose= 2.62336 -0.762426 2.66271

m_count 23

Average Scan Matching Score=673.512

neff= 23.2789

Registering Scans:Done

update frame 3209

update Id=0.419735 ad=0.506211

Laser Pose= 2.5988 -0.723735 2.39106

m_count 24

Average Scan Matching Score=674.315

neff= 23.1565

Registering Scans:Done

Warning: TF_REPEATED_DATA ignoring data with redundant timestamp for frame

arm_left_7_target (parent odom_combined) at time 647.882000 according to authority

unknown_publisher

at line 277 in /tmp/binarydeb/ros-noetic-tf2-0.7.10/src/buffer_core.cpp

Warning: TF_REPEATED_DATA ignoring data with redundant timestamp for frame
arm_right_7_target (parent odom_combined) at time 647.882000 according to authority
unknown_publisher

at line 277 in /tmp/binarydeb/ros-noetic-tf2-0.7.10/src/buffer_core.cpp

update frame 4325

update Id=0.517996 ad=0.509366

Laser Pose= 2.26372 -0.331127 2.42427

m_count 25

Average Scan Matching Score=679.004

neff= 23.0365

Registering Scans:Done

update frame 4340

update Id=0.507391 ad=0.54149

Laser Pose= 1.8136 -0.109964 2.96576

m_count 26

Average Scan Matching Score=654.604

neff= 23.2244

Registering Scans:Done

update frame 4347

update Id=0.338459 ad=0.553547

Laser Pose= 1.48041 -0.136907 -2.76388

m_count 27

Average Scan Matching Score=641.994

neff= 22.7936

Registering Scans:Done

update frame 4354

update Id=0.291325 ad=0.579018

Laser Pose= 1.25436 -0.314296 -2.18486

m_count 28

Average Scan Matching Score=667.966

neff= 22.9504

Registering Scans:Done

update frame 4360

update Id=0.25038 ad=0.503779

Laser Pose= 1.16674 -0.546118 -1.68108

m_count 29

Average Scan Matching Score=653.749

neff= 23.0987

Registering Scans:Done

update frame 4394

update Id=0.19659 ad=0.500007

Laser Pose= 1.18739 -0.738928 -1.18174

m_count 30

Average Scan Matching Score=666.554

neff= 22.9856

Registering Scans:Done

^Cvirtual GMapping::GridSlamProcessor::~GridSlamProcessor(): Start

virtual GMapping::GridSlamProcessor::~GridSlamProcessor(): Deleting tree

labuser@lab-ubuntu-01:~\$

labuser@lab-ubuntu-01:~\$ rosrun gmapping slam_gmapping \

> scan:=/scan_unified \

> _base_frame:=base_link \

> _odom_frame:=odom_combined \

> __log_level:=debug

Warning: TF_REPEATED_DATA ignoring data with redundant timestamp for frame

odom_combined (parent map) at time 770.187000 according to authority unknown_publisher

at line 277 in /tmp/binarydeb/ros-noetic-tf2-0.7.10/src/buffer_core.cpp

[INFO] [1783023525.463677491, 770.177000000]: Laser is mounted upwards.

-maxUrange 29.49 -maxUrange 29.49 -sigma 0.05 -kernelSize 1 -lstep 0.05 -lobsGain 3

-astep 0.05

-srr 0.1 -srt 0.2 -str 0.1 -stt 0.2

-linearUpdate 1 -angularUpdate 0.5 -resampleThreshold 0.5

-xmin -100 -xmax 100 -ymin -100 -ymax 100 -delta 0.05 -particles 30

[INFO] [1783023525.472784307, 770.183000000]: Initialization complete

update frame 0

update ld=0 ad=0

Laser Pose= 1.18798 -0.738316 -1.1843

m_count 0

Registering First Scan

update frame 100

update ld=0.488187 ad=0.504232

Laser Pose= 1.24607 -0.868087 -1.03749

m_count 1

Average Scan Matching Score=648.173

neff= 29.9795

Registering Scans:Done

update frame 216

update ld=1.00322 ad=0.413164

Laser Pose= 1.80081 -1.70261 -1.00556

m_count 2

Average Scan Matching Score=644.67

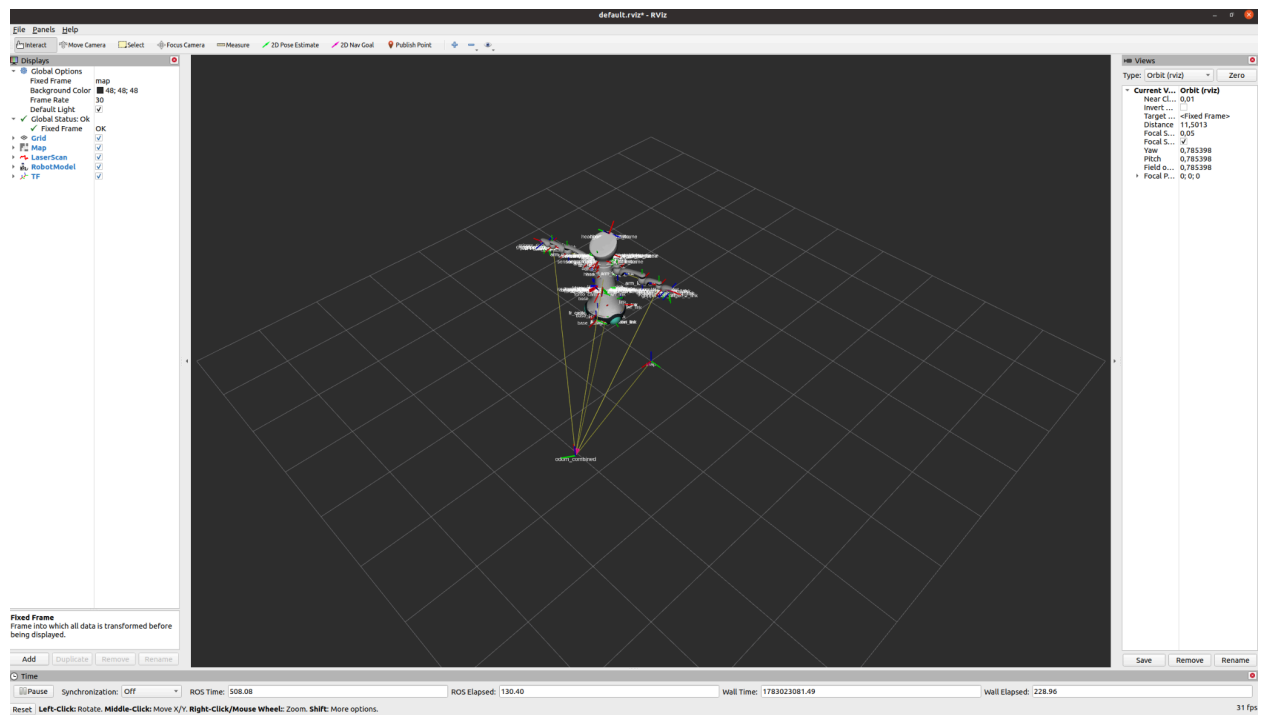
neff= 29.71

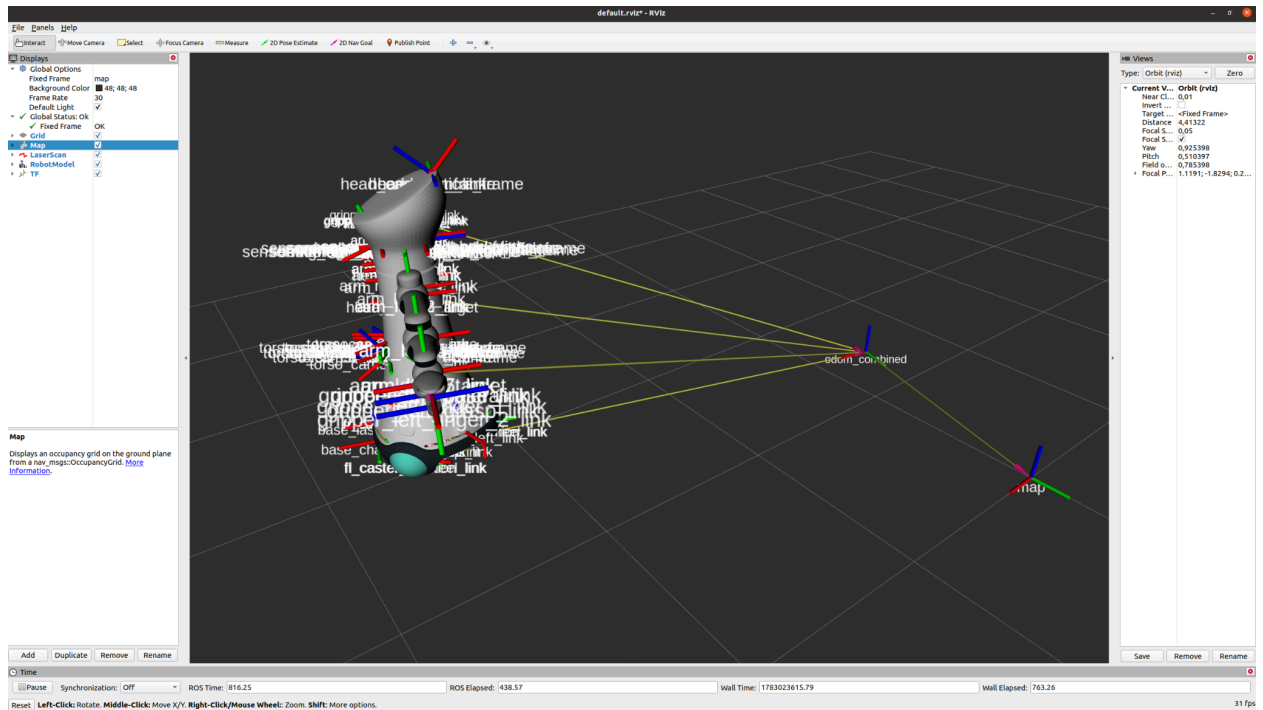
Registering Scans:Done
update frame 241
update Id=0.563054 ad=0.503101
Laser Pose= 2.18492 -2.10468 -0.502496
m_count 3
Average Scan Matching Score=645.838
neff= 29.7075
Registering Scans:Done
update frame 249
update Id=0.338382 ad=0.561521
Laser Pose= 2.50918 -2.18485 0.0590249
m_count 4
Average Scan Matching Score=337.723
neff= 29.6163
Registering Scans:Done
update frame 255
update Id=0.252301 ad=0.500869
Laser Pose= 2.74719 -2.10924 0.559894
m_count 5
Average Scan Matching Score=623.437
neff= 26.6546
Registering Scans:Done
update frame 261
update Id=0.251925 ad=0.503923
Laser Pose= 2.91886 -1.92842 1.06382
m_count 6
Average Scan Matching Score=635.548
neff= 24.7885
Registering Scans:Done
update frame 267
update Id=0.251917 ad=0.503996
Laser Pose= 2.98187 -1.68718 1.56781
m_count 7
Average Scan Matching Score=646.378
neff= 24.7261
Registering Scans:Done
update frame 274
update Id=0.268514 ad=0.548669
Laser Pose= 2.91259 -1.43097 2.11648
m_count 8
Average Scan Matching Score=634.976
neff= 22.7482
Registering Scans:Done
update frame 320

update ld=0.543832 ad=0.506356
Laser Pose= 2.54497 -1.035 2.15391
m_count 9
Average Scan Matching Score=507.041
neff= 13.7121
*****RESAMPLE*****
Deleting Nodes: 0 1 5 7 8 9 10 12 13 14 15 20 21 24 26 29 Done
Deleting old particles...Done
Copying Particles and Registering scans... Done
update frame 331
update ld=0.455403 ad=0.551982
Laser Pose= 2.39692 -0.610197 1.60193
m_count 10
Average Scan Matching Score=519.737
neff= 23
Registering Scans:Done
update frame 338
update ld=0.296603 ad=0.549561
Laser Pose= 2.46402 -0.325117 1.05237
m_count 11
Average Scan Matching Score=611.064
neff= 21.9999
Registering Scans:Done
update frame 344
update ld=0.251508 ad=0.502269
Laser Pose= 2.63708 -0.14617 0.550102
m_count 12
Average Scan Matching Score=584.027
neff= 19.9908
Registering Scans:Done
update frame 352
update ld=0.237297 ad=0.515216
Laser Pose= 2.86073 -0.0751854 0.0348853
m_count 13
Average Scan Matching Score=598.942
neff= 19.9823
Registering Scans:Done

```
roslaunch gmapping slam_gmapping \
  scan:=/scan_unified \
  _base_frame:=base_link \
  _odom_frame:=odom_combined
```

```
roslaunch gmapping slam_gmapping \
  scan:=/scan_unified \
  _base_frame:=base_link \
  _odom_frame:=odom_combined \
  __log_level:=debug
```





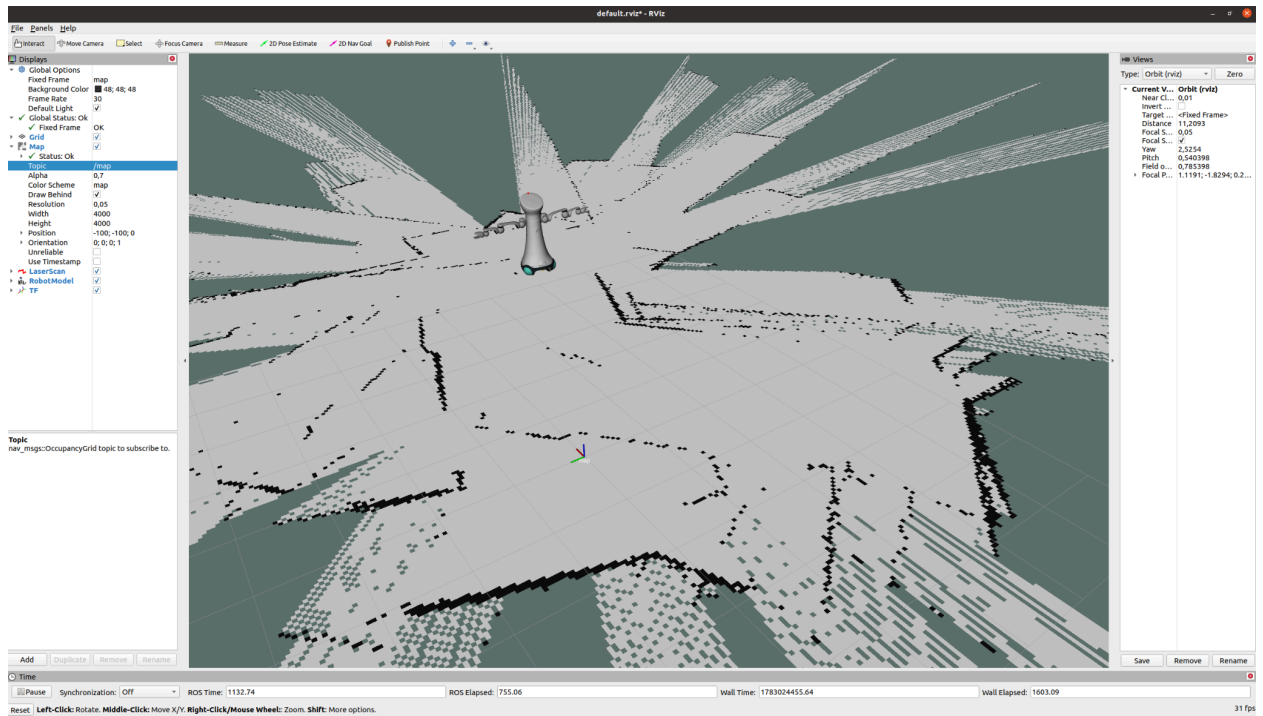
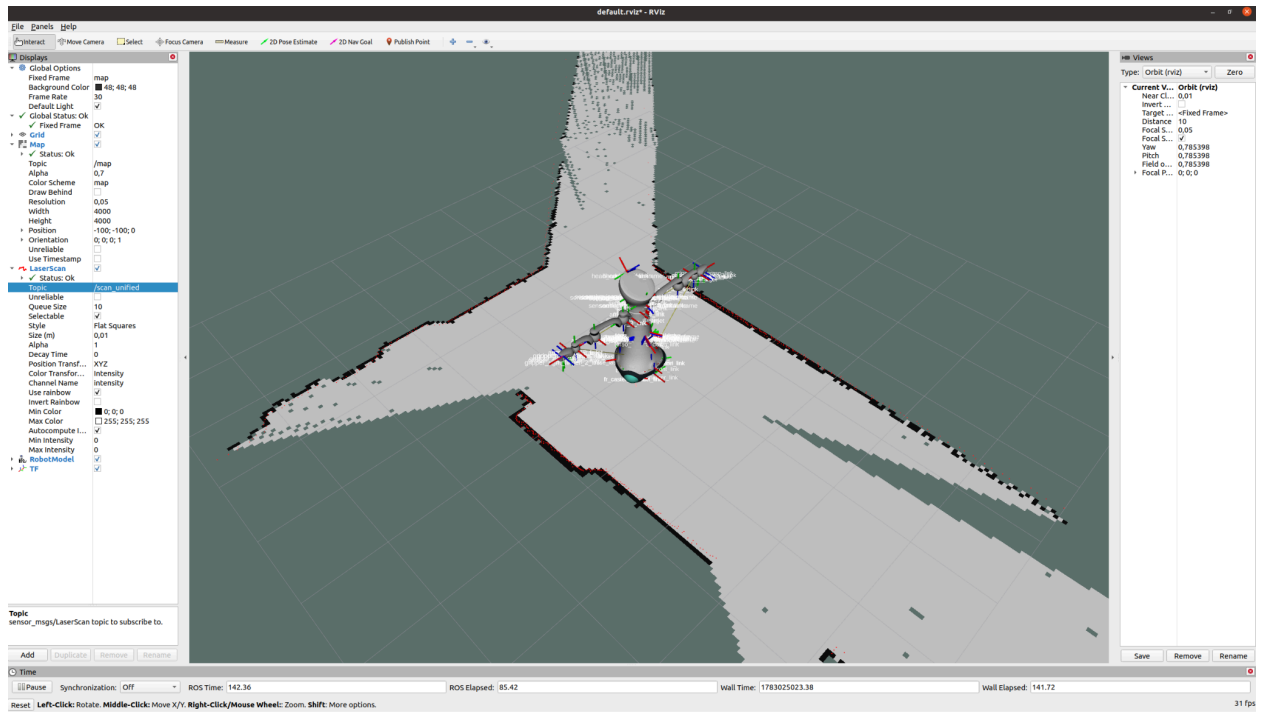
Set:

- **Fixed Frame:** `map`

Add:

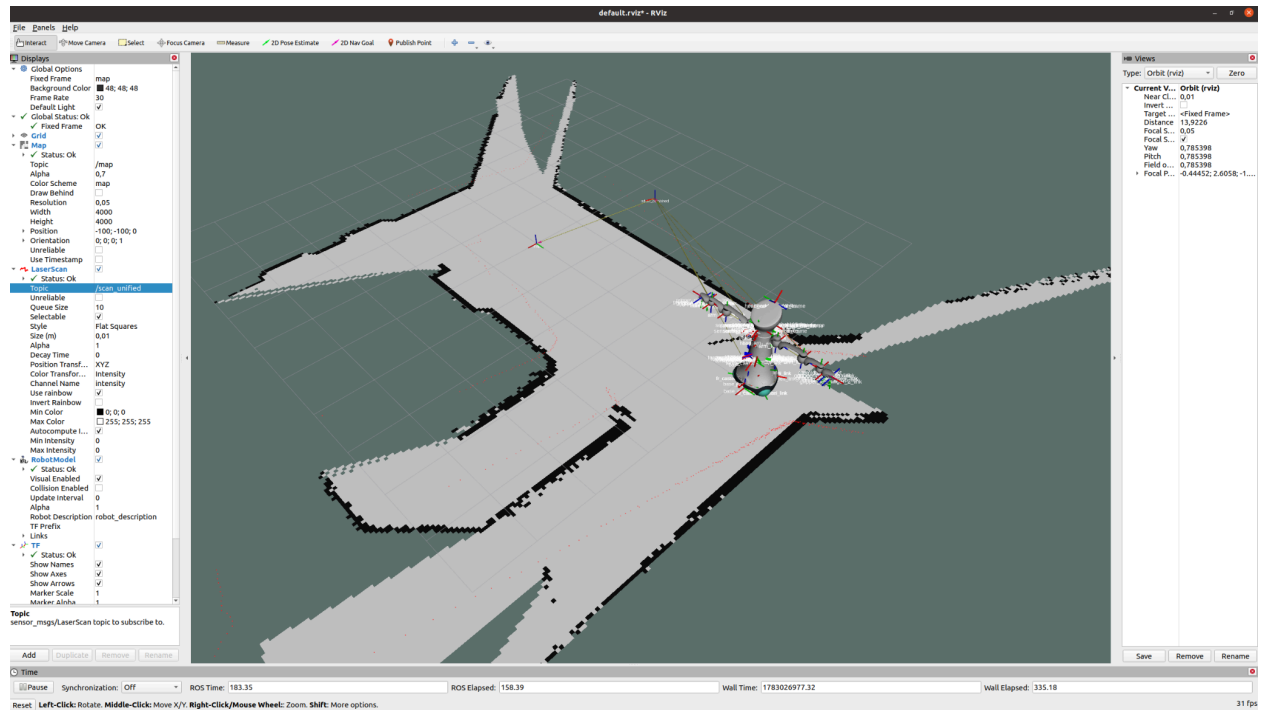
- **Map**
- **LaserScan** (topic: `/scan_unified`)
- **RobotModel**
- **TF**



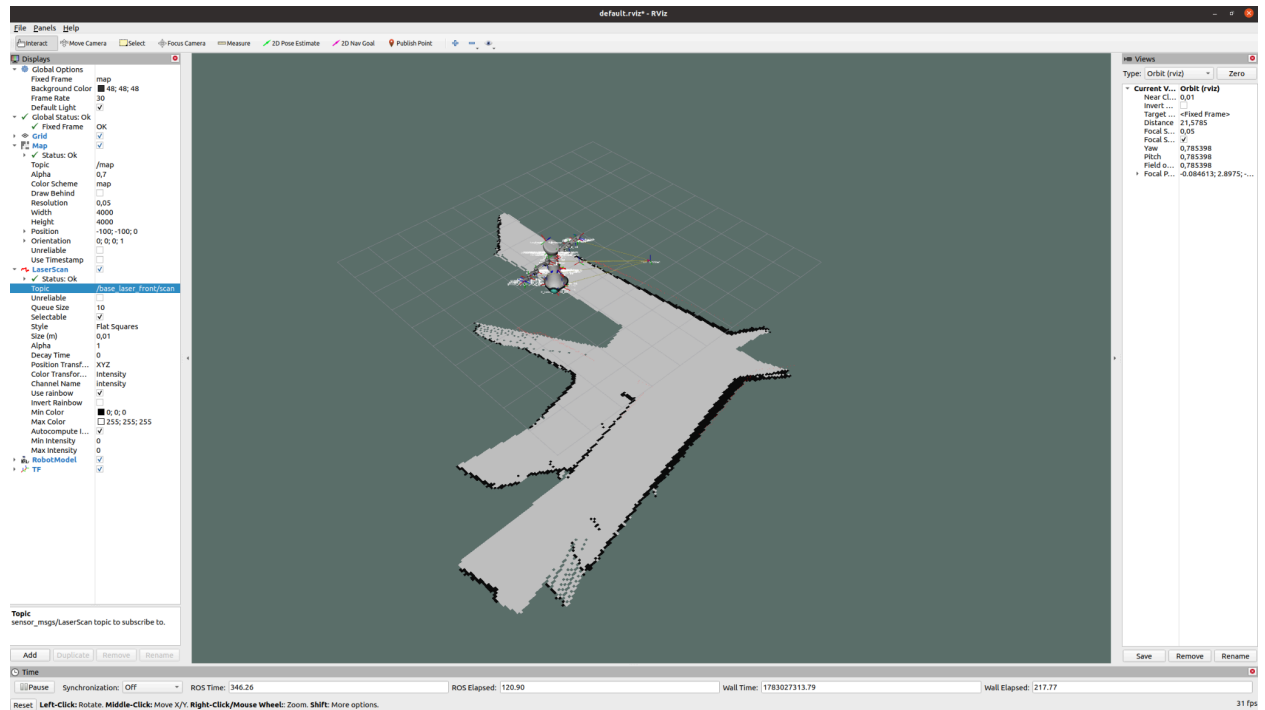


```
labuser@lab-ubuntu-01:~$ rostopic echo -n 1 /scan_unified | grep angle
angle_min: -3.141505479812622
angle_max: 3.141505479812622
angle_increment: 0.008726666681468487
labuser@lab-ubuntu-01:~$ rostopic echo -n 1 /scan_unified | grep range_max
range_max: 29.5
labuser@lab-ubuntu-01:~$ rostopic list | grep scan
/base_laser_front/scan
/base_laser_front/scan_raw
/base_laser_left/scan
/base_laser_left/scan_raw
/base_laser_right/scan
/base_laser_right/scan_raw
/scan_unified
/scan_unified_filtered
/scan_unified_throttled
/scan_unifier_filter/range/parameter_descriptions
/scan_unifier_filter/range/parameter_updates
/station_detector/scan
labuser@lab-ubuntu-01:~$ rostopic echo -n 1 /scan_unified/header
seq: 9280
stamp:
  secs: 777
  nsecs: 51000000
frame_id: "base_link"
---
```

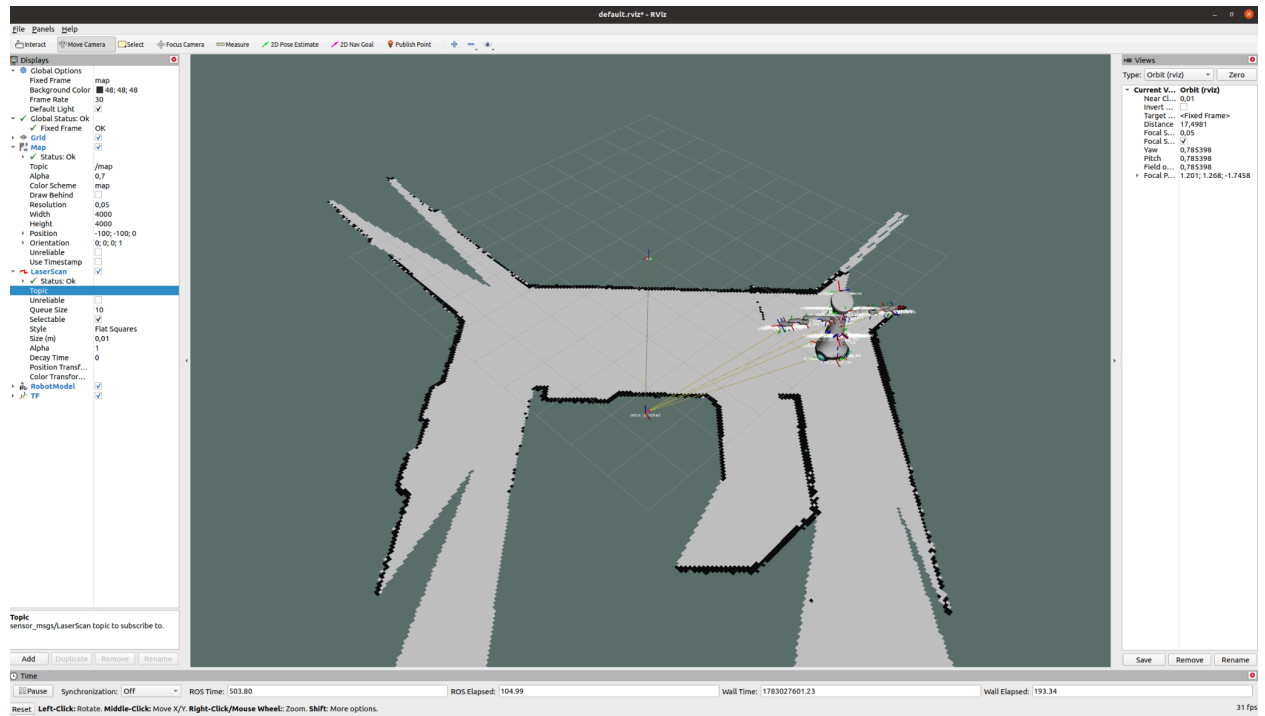
```
labuser@lab-ubuntu-01:~$
```



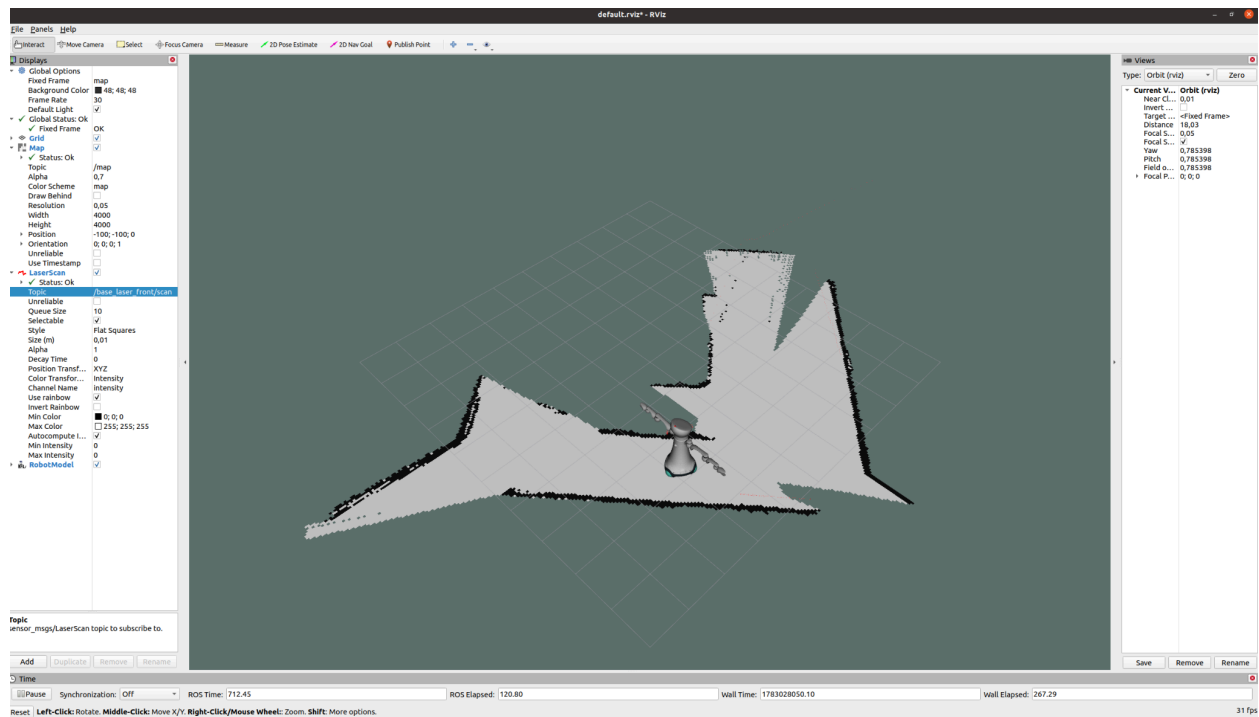
```
roslaunch gmapping slam_gmapping \
  scan:=/base_laser_front/scan \
  _odom_frame:=odom_combined
```



```
roslaunch gmapping slam_gmapping \
  scan:=/base_laser_front/scan \
  _odom_frame:=odom_combined \
  _maxRange:=8 \
  _maxUrange:=7.5 \
  _linearUpdate:=0.2 \
  _angularUpdate:=0.1 \
  _particles:=80
```

```
roslaunch gmapping slam_gmapping \
  scan:=/base_laser_front/scan \
  _odom_frame:=odom_combined \
  _maxRange:=8 \
  _maxUrange:=7.5 \
  _linearUpdate:=0.2 \
  _angularUpdate:=0.1 \
  _particles:=80
```



```
labuser@lab-ubuntu-01:~$ rostopic echo -n 1 /base_laser_front/scan/header
seq: 8620
stamp:
  secs: 724
  nsecs: 736000000
frame_id: "base_laser_front_link"
```

```
labuser@lab-ubuntu-01:~$ rosrn tf tf_echo base_link base_laser_front_link
Warning: TF_REPEATED_DATA ignoring data with redundant timestamp for frame
arm_right_7_target (parent odom_combined) at time 726.951000 according to authority
unknown_publisher
at line 277 in /tmp/binarydeb/ros-noetic-tf2-0.7.10/src/buffer_core.cpp
```


[illegible]

0.6586798429489136, 0.6976583003997803, 0.7032878994941711, 0.7056966423988342,
0.0,
0.0,
0.0,
0.0,
0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0]

[illegible]

— — —

labuser@lab-ubuntu-01:~\$

```
^Clabuser@lab-ubuntu-01:~$ rosrn tf tf_echo map odom combined
```

At time 778.744

- Translation: [8.361, 1.733, 0.000]

- Rotation: in Quaternion [0.000, 0.000, 0.843, 0.538]

in RPY (radian) [0.000, -0.000, 2.007]

in RPY (degree) [0.000, -0.000, 114.969]

At time 778.744

- Translation: [8.361, 1.733, 0.000]

- Rotation: in Quaternions [0.000, 0.000, 0.843, 0.538]

```
    in RPY (radian) [0.000, -0.000, 2.007]
    in RPY (degree) [0.000, -0.000, 114.969]
At time 779.744
- Translation: [8.361, 1.733, 0.000]
- Rotation: in Quaternion [0.000, 0.000, 0.843, 0.538]
           in RPY (radian) [0.000, -0.000, 2.007]
           in RPY (degree) [0.000, -0.000, 114.969]
^Clabuser@lab-ubuntu-01:~$
labuser@lab-ubuntu-01:~$
labuser@lab-ubuntu-01:~$
labuser@lab-ubuntu-01:~$
labuser@lab-ubuntu-01:~$
labuser@lab-ubuntu-01:~$
labuser@lab-ubuntu-01:~$
labuser@lab-ubuntu-01:~$
labuser@lab-ubuntu-01:~$
labuser@lab-ubuntu-01:~$ rostopic echo -n 1 /base_laser_front/scan | grep angle
angle_min: -2.3561999797821045
angle_max: 2.3561999797821045
angle_increment: 0.008726666681468487
labuser@lab-ubuntu-01:~$ rospack list | grep tiago
labuser@lab-ubuntu-01:~$ echo $ROS_PACKAGE_PATH
/opt/ros/noetic/share
labuser@lab-ubuntu-01:~$
labuser@lab-ubuntu-01:~$ rostopic list | grep odom
/base/odometry_controller/odometry
labuser@lab-ubuntu-01:~$ rostopic list | grep wheel
/base/twist_controller/wheel_commands
labuser@lab-ubuntu-01:~$
```